

SUMMARY

Aspiring software engineer specialising in **gameplay and engine development and large-scale systems**. Skilled in C++ and Python, with hands-on experience in graphics, AI pipelines, and gameplay tools. Passionate about building scalable, player-focused technologies and immersive game experiences.

EDUCATION

Southern Methodist University <i>Master of Interactive Technology in Digital Game Development (Software Development)</i>	Dallas, United States <i>Aug 2025 – Present</i>
Technical University of Munich <i>Exchange Student, Bachelor of Informatics</i>	Munich, Germany <i>Apr 2024 – Aug 2024</i>
National University of Singapore <i>Bachelor of Computing in Computer Science (Honours)</i>	Singapore <i>Aug 2021 – June 2025</i>
◦ Specialisation: Computer Graphics and Games, Artificial Intelligence	

PUBLICATIONS

- **Sun, R.**, Li, T., Tim, T., Ng, Q., Bhojan, A. (2025). GAGA: Game Asset Generative AI Pipeline. **International Conference on Teaching, Assessment and Learning for Engineering (IEEE TALE 2025)**.
- Janaka, N., Zhao, S., Ram, A., **Sun, R.**, Tan Jing Wen, S., Li, D., & Hsu, D. (2025). Progressive Sentences: Combining the Benefits of Word and Sentence Learning. **Adjunct Proceedings of the 27th International Conference on Mobile Human-Computer Interaction, 1–12**. <https://doi.org/10.1145/3737821.3749564>

EXPERIENCE

Fragcolor Pte. Ltd. <i>Game Engine Programmer Intern</i>	Singapore <i>May 2023 – Oct 2023</i>
◦ Developed and optimised C++ engine modules , improving rendering scalability and runtime efficiency.	
◦ Implemented gameplay and visual systems for the company’s flagship title to demonstrate engine capabilities.	
◦ Prototyped LLM-powered no-code workflows , reducing iteration time for designers and non-technical users.	
◦ Collaborated with engineers in an agile environment to improve performance, stability, and usability.	
The Synteraction (formerly NUS-HCI) Lab <i>Part-time Research Intern</i>	Singapore <i>Feb 2024 – Dec 2024</i>
◦ Conducted literature review on multimedia learning systems, building a foundation for new research directions.	
◦ Designed and led usability experiments , producing actionable insights that improved interaction flow.	
◦ Developed interactive prototypes using Python and React , evaluated through iterative user testing.	

PROJECTS

MetalCore (Unreal Engine)
◦ Best Project Award in the 22nd STePS NUS School of Computing Term Project Showcase .
◦ Built gameplay systems and UI in Unreal Engine using C++/Blueprints, enhancing player engagement.
◦ Collaborated with a team of 5 to design combat, progression, and core gameplay loops .
Glimmerwood (GameMaker Studio)
◦ Created a side-scrolling endless runner with player skills, interactive bosses, and scripted events.
◦ Facilitated playtesting and integrated feedback, improving balance and gameplay clarity.
3D Object Reconstruction Pipeline (Final Year Project)
◦ Developed a modular Python pipeline to generate PBR-textured 3D assets from 2D images.
◦ Containerised with Docker and deployed via REST APIs, built a web viewer in React and Three.js .
◦ Benchmarked performance across 50+ object types, optimising GPU runtime with CUDA.

SKILLS

Programming: Python, Java, C, C++, C#, R, Shell Scripting, PostgreSQL, OpenGL
Engines: Unity3D, Unreal Engine, GameMaker
Tools: Git, Perforce, Blender, Figma, Photoshop, Illustrator